



Final Report – “Mountain River Scroll”

Progress and Achievements

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06

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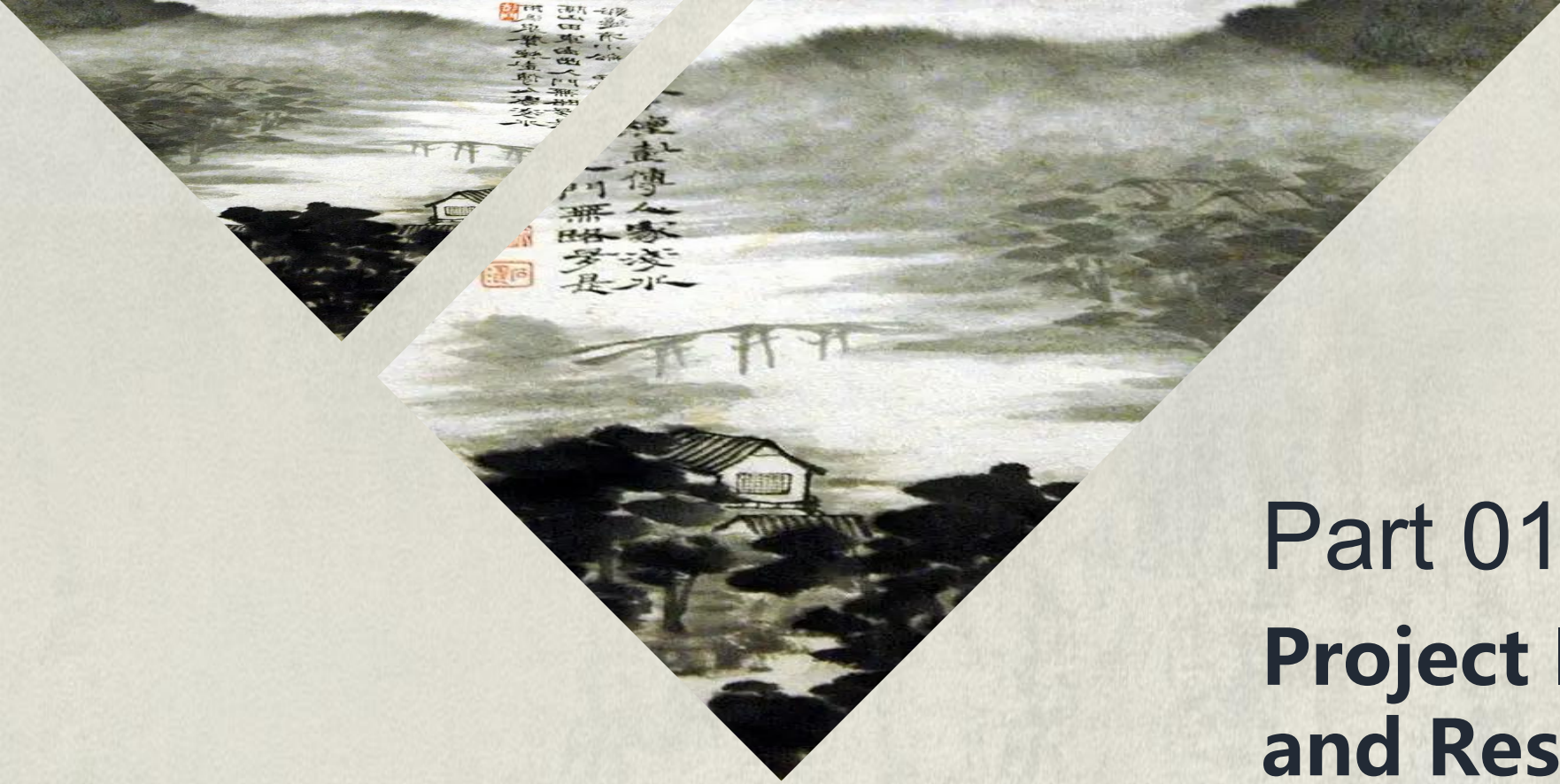
3.

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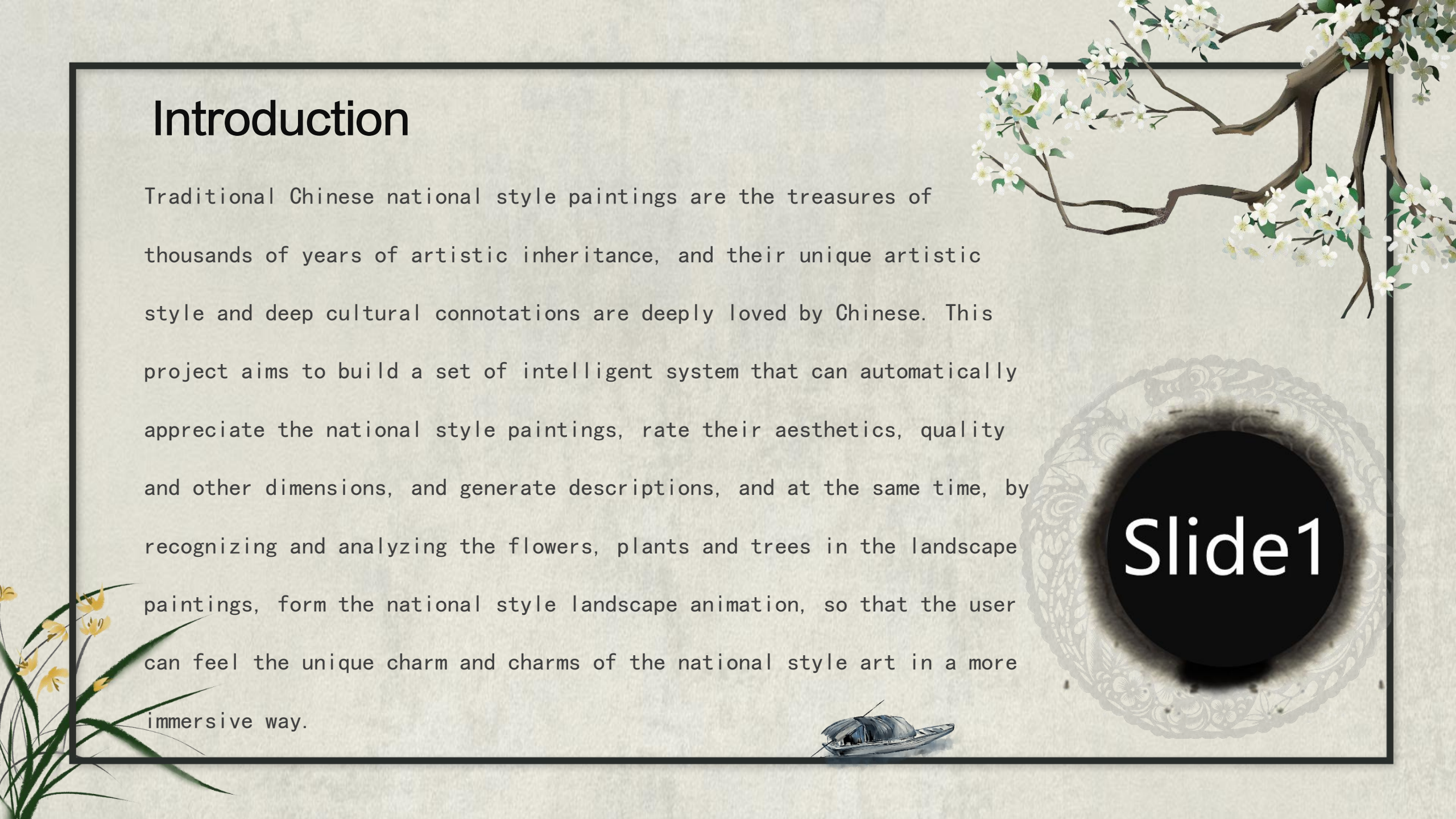


Part 01

Project Introduction and Research Background

Introduction

Traditional Chinese national style paintings are the treasures of thousands of years of artistic inheritance, and their unique artistic style and deep cultural connotations are deeply loved by Chinese. This project aims to build a set of intelligent system that can automatically appreciate the national style paintings, rate their aesthetics, quality and other dimensions, and generate descriptions, and at the same time, by recognizing and analyzing the flowers, plants and trees in the landscape paintings, form the national style landscape animation, so that the user can feel the unique charm and charms of the national style art in a more immersive way.



Slide1

Functional Technologies

➤ MiniGPT-4

Use of visual features aligned with advanced LLMs to obtain results similar to GPT at the CV technology level

➤ Image-to-Video

Image-to-Video is a video diffusion model (VLDM) based on latent space.

➤ Reward CLIP+MLP Aesthetic Scoring

Aesthetic evaluation of images based on a simple neural network with CLIP embeddings as inputs

➤ ModelScope

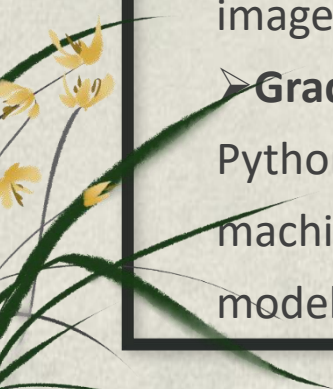
ModelScope favors end-to-end implementation of the model, using QWEN and image loss assessment in a simple way

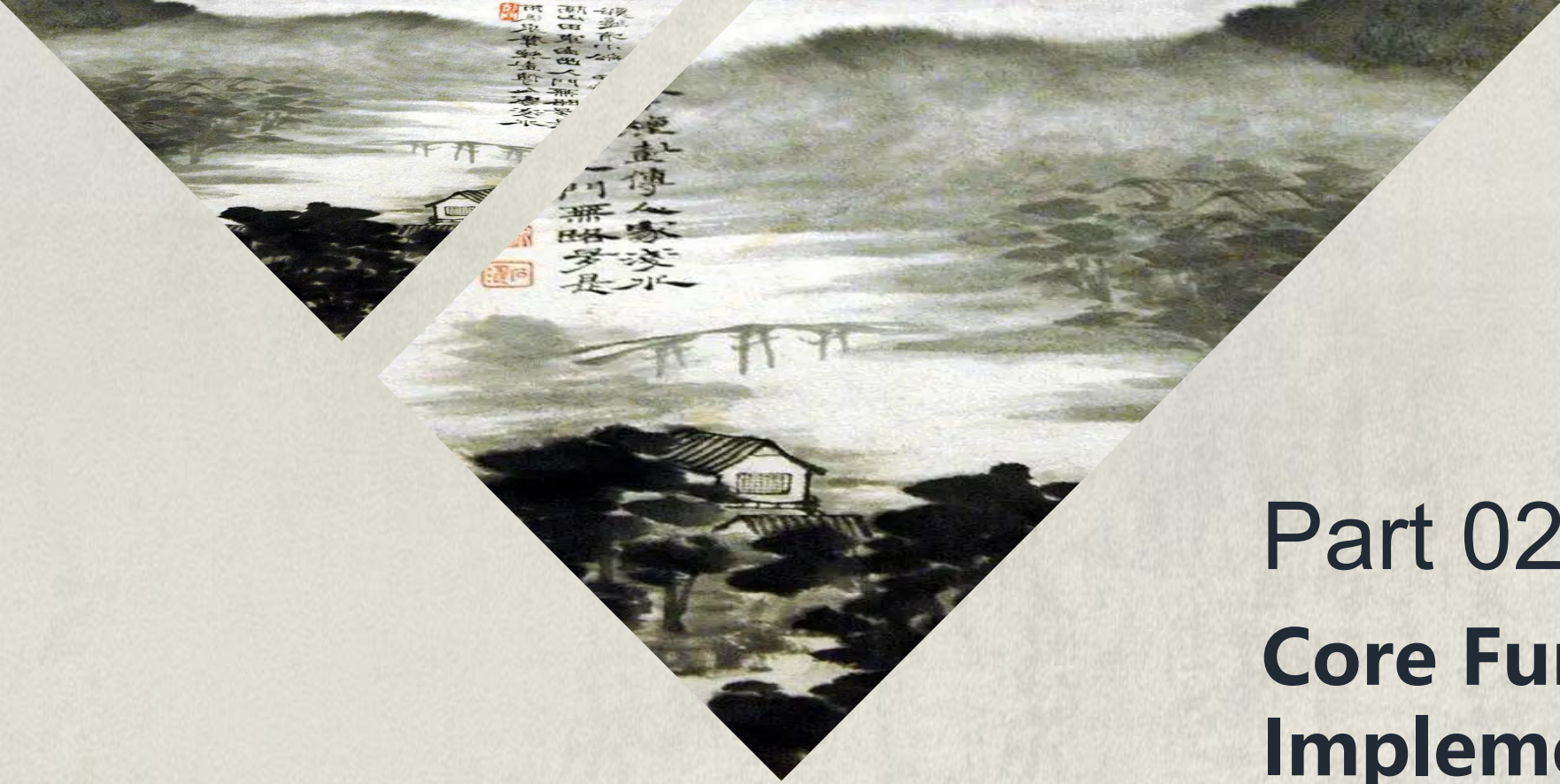
➤ Gradio

Python library for building interactive interfaces for presenting and testing machine learning models, natural language processing models, computer vision models



Slide2





Part 02

Core Functions Implementation



Image Description



MiniGPT-4:

Enhancing Vision-language Understanding with Advanced Large Language Models

Deyao Zhu^{*}, Jun Chen^{*}, Xiaoqian Shen, Xiang Li, Mohamed Elhoseiny

► King Abdullah University of Science and Technology

^{*}Equal Contribution

✕ Paper

🔄 Code

😊 Space

📺 Video

📄 Dataset

😊 Model

MiniGPT-4

AI image content analysis,
which can be used for logical
captcha inference recognition

OFA

OFA unifies various cross modal and single modal tasks into a simple sequence to sequence learning framework, including image generation, image classification, language modeling, etc

OFA的轻量化

本模型是在对OFA的large模型版本基础上通过知识蒸馏进行轻量化而得到的tiny版本模型（参数量33M），方便用户部署在存储和计算资源受限的设备上。蒸馏框架介绍，详见论文：[Knowledge Distillation of Transformer-based Language Models Revisited](#)，以及我们的官方Github仓库<https://github.com/OFA-Sys/OFA-Compress>

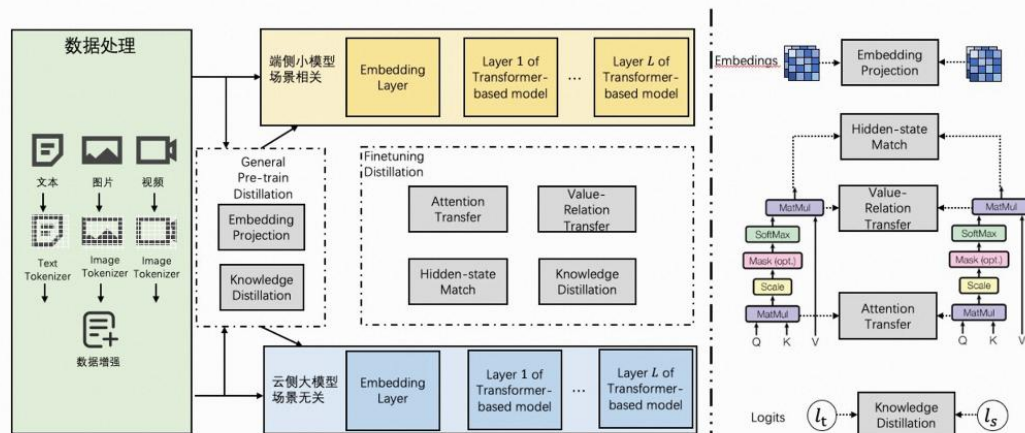




Image Description



MiniGPT-4

Open source large model published in 2023.11. Using visual features aligned with advanced large language models to obtain results similar to GPT at the CV technology level. We carry out the deployment of MiniGPT-4 in linux server and use its image analysis function to realize the image description functions. And use our own dataset of ancient style images for fine-tuning training.

MINIGPT-4: ENHANCING VISION-LANGUAGE UNDERSTANDING WITH ADVANCED LARGE LANGUAGE MODELS

Deyao Zhu*, Jun Chen*, Xiaoqian Shen, Xiang Li, Mohamed Elhoseiny
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Train

The main core of minigpt4's alignment of the visual coder with LLM is the use of 2-stage model research and deployment when training Linear. There are two phases of training. The first stage is training on a dataset for language comprehension. The second stage allows training on a dataset using a small range of image-text. We used the dataset we collected and cleaned for the second stage of fine-tuned training.

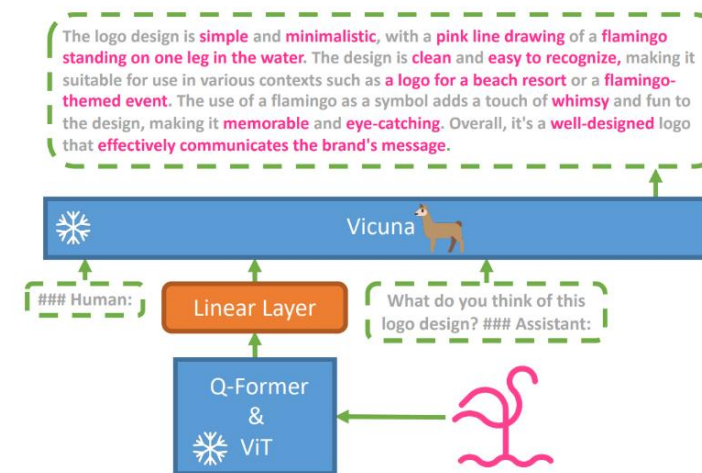


Figure 1: **The architecture of MiniGPT-4.** It consists of a vision encoder with a pretrained ViT and Q-Former, a single linear projection layer, and an advanced Vicuna large language model. MiniGPT-4 only requires training the linear projection layer to align the visual features with the Vicuna.



Image Quality Inspection and Q&A Module



Modelscope

In order to reduce the burden of the whole system and improve the efficiency. In the quality detection and natural language part, the modelscope framework is used to realize this part by calling the model directly. By using the image damage model provided by modelscope, the quality of the image is detected. The quality score of the image is graded and evaluated as "good, excellent

```
def qualitycheck(imgPath):  
    image_quality_assessment_degradation_pipeline = pipeline(Tasks.image_quality_assessment_degradation, 'damo/cv_resnet50_image-qual  
    result = image_quality_assessment_degradation_pipeline(imgPath)  
    dictResult = dict(zip(result[OutputKeys.LABELS], result[OutputKeys.SCORES]))  
    zaosheng = float(dictResult['噪声强度'])  
    mohu = float(dictResult['模糊程度'])  
    yasuo = float(dictResult['压缩强度'])
```

QWen

By using the channels provided by modelscope to call the natural language model, the translation of a thousand questions, using the output text of MiniGPT-4 as input, and by fixing the prompt to get the corresponding stable results.

```
# Note: The default behavior now has injection attack prevention off.  
model  
tokenizer  
def initial():  
    global tokenizer  
    tokenizer = AutoTokenizer.from_pretrained("qwen/Qwen-14B-Chat", trust_remote_code=True)  
    global model  
    model = AutoModelForCausalLM.from_pretrained("qwen/Qwen-14B-Chat", device_map="auto", trust  
# use bf16  
# model = AutoModelForCausalLM.from_pretrained("qwen/Qwen-14B-Chat", device_map="auto", trust
```



TONGYI
Qwen

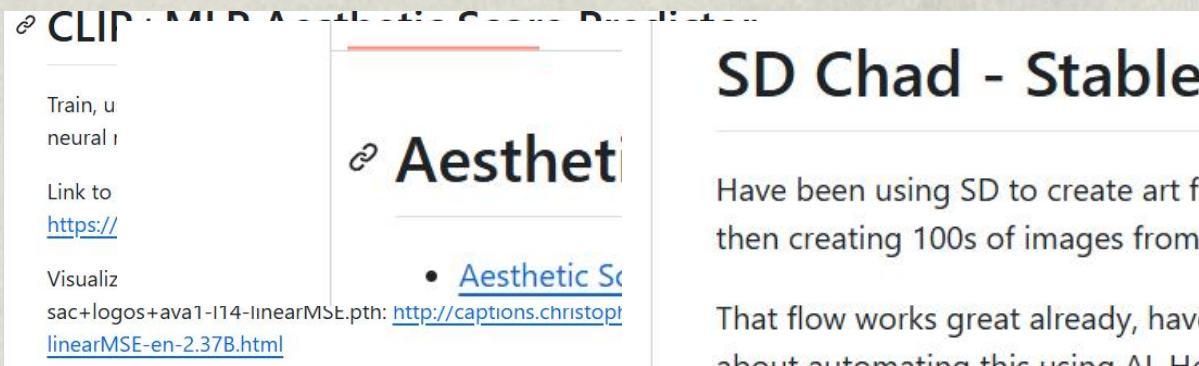


Aesthetic evaluation



Aestgetic

The use of image rewards (human preference algorithm) and aesthetic score composed of CLIP+BLM during dataset processing The aesthetic evaluation algorithm uses CLIP's image feature extraction as input, connected to a neural network model for evaluation



```
class AestheticPredictor(nn.Module):
    def __init__(self, input_size: int):
        super().__init__()
        self.input_size = input_size
        self.layers = nn.Sequential(
            nn.Linear(self.input_size, 1024),
            nn.Dropout(0.2),
            nn.Linear(1024, 128),
            nn.Dropout(0.2),
            nn.Linear(128, 64),
            nn.Dropout(0.1),
            nn.Linear(64, 16),
            nn.Linear(16, 1),
```

Image-Reward

is a modeling algorithm for evaluating human preferences, which is currently more commonly used in text-to-image generation

ImageReward: Learning and Evaluating Human Preferences for Text-to-Image Generation

Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, Yuxiao Dong

We present a comprehensive solution to learn and improve text-to-image models from human preference feedback. We propose a human preference reward model -- to effectively encode human preferences. Its training is based on our system's human preference comparisons to date. In human evaluation, ImageReward outperforms existing scoring models and metrics, making it, we propose Reward Feedback Learning (ReFL), a direct tuning algorithm to optimize diffusion models against human preference compared methods. All code and datasets are provided at [this https URL](https://github.com/IDR-AILab/ImageReward).

```
def RewardFunc(imgPath):
    model = RM.load("ImageReward-v1.0")

    rewards = model.score("landscape painting", [imgPath])
    resultFloat = float(rewards)
```

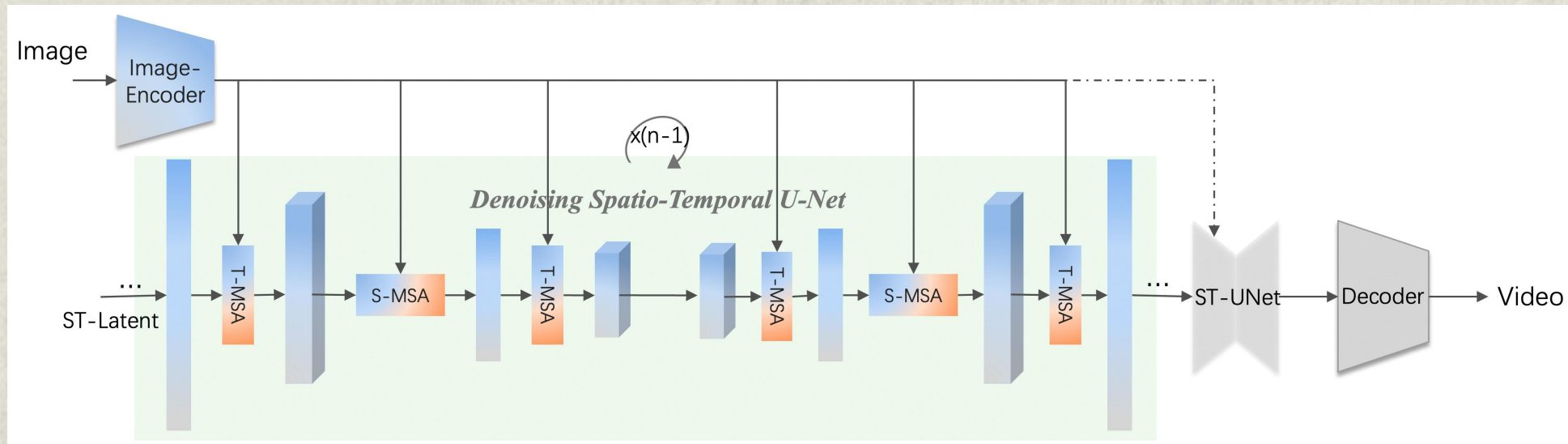



Video Generation



Image-to-Video

Image-to-Video is a video latent diffusion model. It utilizes our designed ST-UNet to perform spatio-temporal modeling in the latent space and reconstruct the generated video through a decoder. In order to generate 720P videos, Image-to-Video was into two stages. The first stage ensures semantic consistency with low resolutions, while the second stage utilizes the new VLDM to denoise and improve video resolution, as well as enhance temporal and spatial consistency.





Video Generation



First Stage

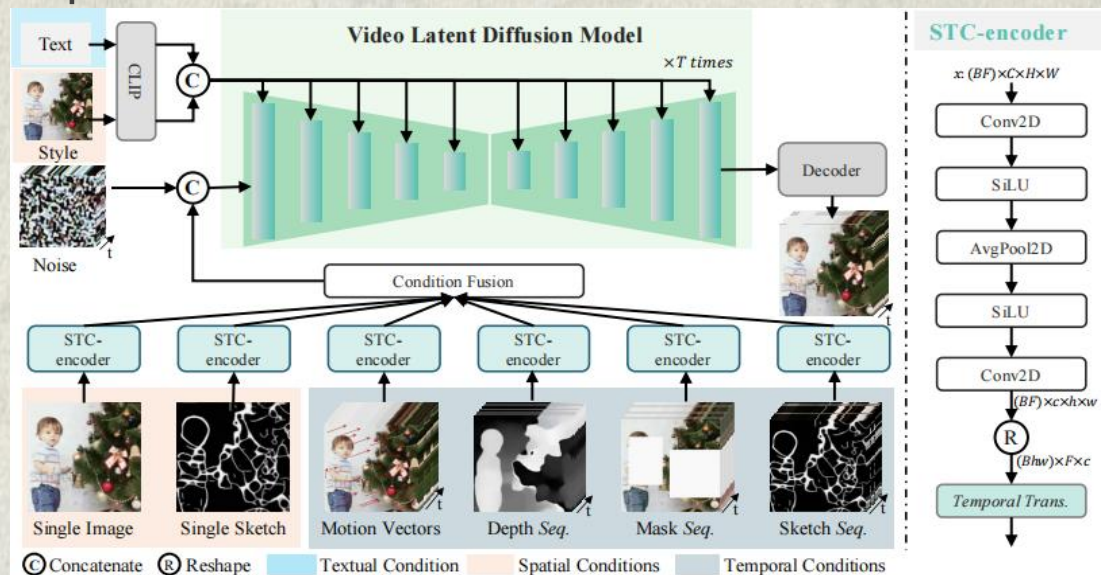
- The 3D-UNet architecture, denoted as $\epsilon_{\theta}(\cdot, \cdot, t)$, effectively addresses the issue of semantic consistency through a combination of spatial and temporal modeling. This is achieved by extending the 2D UNet into a 3D UNet, incorporating both spatial and temporal dimensions to ensure a coherent understanding across time series data.

In a single UNet block, the following four basic building blocks are used:

- Spatial Convolution
- Temporal Convolution
- Spatial Transformer
- Temporal Transformer

Second Stage

- VLDM is a technique that applies diffusion models to video data. By modeling in latent space, VLDM can effectively process and generate video data. Video data usually contains noise and low resolution. **VLDM not only improves the video resolution through the denoising process, but also maintains consistency in time and space.**





Video Generation



our final effect:

input image:



output short video:





Video Generation



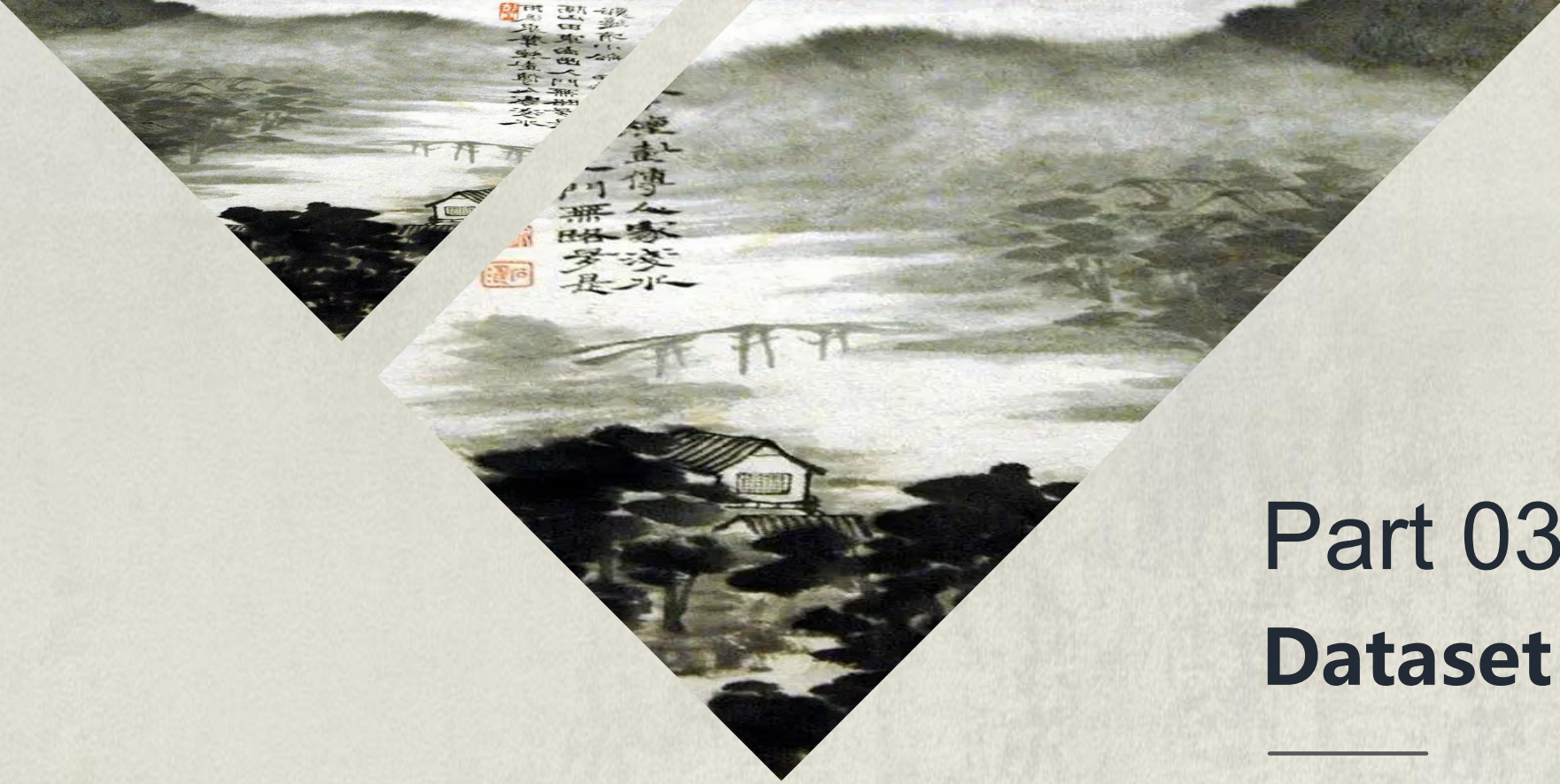
our final effect:

input image:



output short video:





Part 03

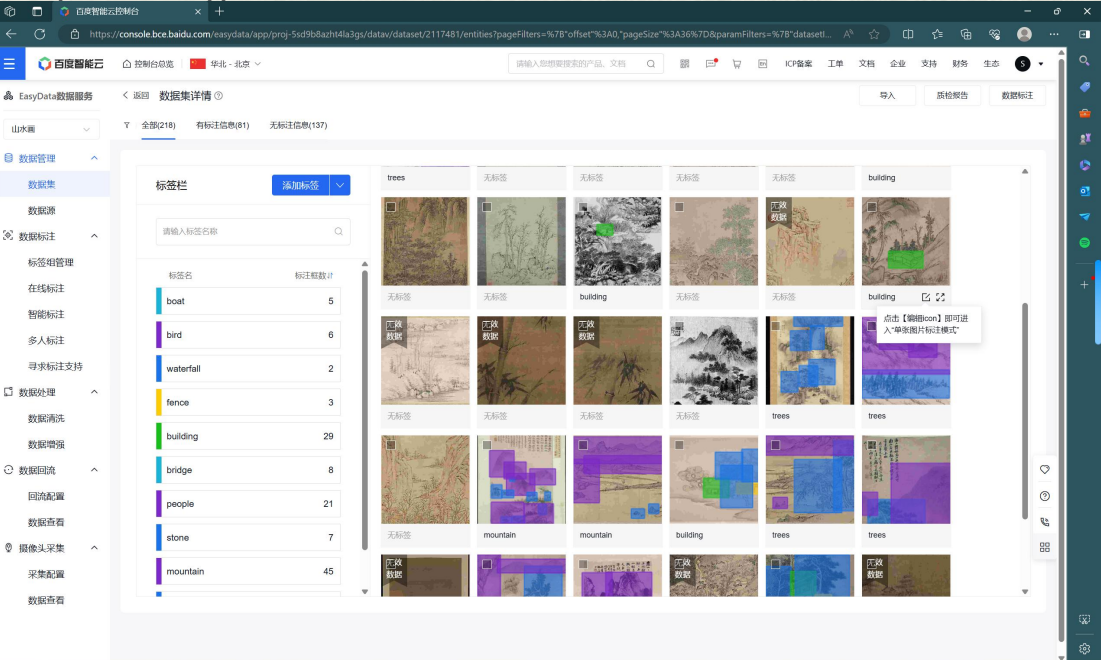
Dataset Processing



Dataset Processing



➤ For the initial dataset, we used Baidu Cloud AI-assisted annotation, and the annotations were uniformly stored using the json format to obtain a roughly processed picture-text dataset of national wind. This part of the dataset includes 1519 pictures of national style landscape paintings and the labeled text.



➤ We have also established a certain number of image quality tests, aesthetic assessments, and datasets of ancient poetic texts

target	filename	image_caption	hed	source
image · width (px)	string · lengths	string · lengths	image · width (px)	image · width (px)
512	11	27	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K
512	11	92	512	62K

id	title	group_index	type	dynamic	author	content
string · lengths	string · lengths	float64	string · lengths	string · lengths	string · lengths	string · lengths
1	0000001.jpg	11	0000001.jpg	11	0000001.jpg	0000001.jpg
2	0000003.jpg	11	0000003.jpg	11	0000003.jpg	0000003.jpg
3	0000006.jpg	11	0000006.jpg	11	0000006.jpg	0000006.jpg
4	0000007.jpg	11	0000007.jpg	11	0000007.jpg	0000007.jpg
5	0000008.jpg	11	0000008.jpg	11	0000008.jpg	0000008.jpg
6	0000009.jpg	11	0000009.jpg	11	0000009.jpg	0000009.jpg
7	0000010.jpg	11	0000010.jpg	11	0000010.jpg	0000010.jpg
8	0000011.jpg	11	0000011.jpg	11	0000011.jpg	0000011.jpg
9	0000012.jpg	11	0000012.jpg	11	0000012.jpg	0000012.jpg
10	0000013.jpg	11	0000013.jpg	11	0000013.jpg	0000013.jpg

Dataset Cleaning

- It is mentioned in the paper that proposing a small number of high quality datasets for training the model gives better results compared to a large number of average quality datasets.

InstructionGPT-4: A 200-Instruction Paradigm for Fine-Tuning MiniGPT-4

Lai Wei, Zihao Jiang, Weiran Huang, Lichao Sun

Multimodal large language models are typically trained in two stages: first pre-training on image-text pairs, and then fine-tuning using supervised vision-language instruction data. Recent studies have shown that large language models can achieve satisfactory results even with a limited amount of high-quality instruction-following data. In this paper, we introduce InstructionGPT-4, which is fine-tuned on a small dataset comprising only 200 examples, amounting to approximately 6% of the instruction-following data used in the alignment dataset for MiniGPT-4. To achieve this, we first propose several metrics to assess the quality of multimodal instruction data. Based on these metrics, we present an effective and trainable data selector to automatically identify and filter low-quality vision-language data. By employing this method, InstructionGPT-4 outperforms the original MiniGPT-4 on various evaluations. Overall, our findings demonstrate that less but high-quality instruction tuning data is efficient in enabling multimodal large language models to generate better output. Our code is available at [this https URL](#).

- In the paper, unsupervised learning is adopted for k-balanced classification, after which each cluster is cleaned using five parameters, CLIP Score, Length Score, Reward Score, GPT Score, and Multimodal Features, as criteria, and the high-quality triples selected from each category are combined to form the final high-quality dataset
- We have reproduced the CLIP score as well as the Reward Score, written classifiers to realize the cleaning of the dataset. And we reintegrated the calculation of these two parameters to apply them to the aesthetic assessment part of the program.





Dataset Cleaning



■ Random Forest Algorithm for Image Style Classification Verification:

```
for label_name, image_paths in images_by_class.items():
    for image_path in image_paths:
        image = cv2.imread(image_path)
        if image is not None:
            image = cv2.resize(image, IMAGE_SIZE)
            if CROP_SIZE is not None:
                h, w = image.shape[:2]
                start_x = (w - CROP_SIZE[1]) // 2
                start_y = (h - CROP_SIZE[0]) // 2
                image = image[start_y:start_y+CROP_SIZE[0], start_x:start_x+CROP_SIZE[1]]

            hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
            h, s, v = cv2.split(hsv)

            # 计算H、S、V通道的颜色直方图
            num_bins = 32 # 增加bin的数量以提高分辨率
            hist_h = cv2.calcHist([h], [0], None, [num_bins], [0, 180])
            hist_h = hist_h.flatten()
            hist_s = cv2.calcHist([s], [0], None, [num_bins], [0, 256])
            hist_s = hist_s.flatten()
            hist_v = cv2.calcHist([v], [0], None, [num_bins], [0, 256])
            hist_v = hist_v.flatten()

            # 合并H、S、V通道的颜色直方图
            hist_color = np.concatenate((hist_h, hist_s, hist_v))

            # 计算灰度共生矩阵的特征
            gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
            glcm = graycomatrix(gray, [1], [0], 256, symmetric=True, normed=True)

            # 分别计算对比度、相关性、能量和同质性
            contrast = graycoprops(glcm, 'contrast')[0, 0]
            correlation = graycoprops(glcm, 'correlation')[0, 0]
            energy = graycoprops(glcm, 'energy')[0, 0]
            homogeneity = graycoprops(glcm, 'homogeneity')[0, 0]

            # 将这些特征放入一个数组中
            glcm_features = np.array([contrast, correlation, energy, homogeneity])

            # 合并特征
            feature_vector = np.concatenate((hist_color, glcm_features))
            feature_vector = normalize(feature_vector.reshape(1, -1), axis=1).flatten()

            # 添加到特征列表和标签列表
            features.append(feature_vector)
            labels.append(label_name)
```

- CLIP+BLM calculates aesthetic assessment scores on images:
- After feature extraction of the image using CLIP, obtain an aesthetic assessment score through a neural network

```
class AestheticPredictor(nn.Module):
    def __init__(self, input_size: int):
        super().__init__()
        self.input_size = input_size
        self.layers = nn.Sequential(
            nn.Linear(self.input_size, 1024),
            nn.Dropout(0.2),
            nn.Linear(1024, 128),
            nn.Dropout(0.2),
            nn.Linear(128, 64),
            nn.Dropout(0.1),
            nn.Linear(64, 16),
            nn.Linear(16, 1),
```

■ Using Image Reward as the Reward score (human preference)

```
def ReWardFunc(imgPath):

    model = RM.load("ImageReward-v1.0")

    rewards = model.score("landscape painting", [imgPath])
    resultFloat = float(rewards)
```

分类算法	SVM	逻辑回归	随机森林	梯度提升	K近邻	决策树
准确率	0.6141	0.5809	0.6597	0.5518	0.6182	0.5394

Finally, 180 high-quality datasets were obtained



Dataset Cleaning



Cleaning

We collected 5,000 video-text pairs from Webvid to form the video-text dataset. The video-text set is accurately described and the video is informative, but there is a watermark that will inevitably affect the training results.

PaddleOCR is an OCR (Optical Character Recognition) system developed based on Flying Paddle. Its technical system includes text detection, text recognition, text direction detection and image processing modules.

The use of PaddleOCR to recognize the text in the video, because there may be other text in the video, so to determine whether it is a watermark part, using the IOU, preset thresholds, if greater than the gate

can be con



```
#计算倾斜矩阵面积
def rectangle_area(rect) -> float:
    #计算向量
    vector1 = [rect[1][0] - rect[0][0], rect[1][1] - rect[0][1]]
    vector2 = [rect[2][0] - rect[0][0], rect[2][1] - rect[0][1]]

    #计算向量的叉积
    cross_product = vector1[0] * vector2[1] - vector1[1] * vector2[0]

    #取叉积的绝对值就是矩形的面积
    area = abs(cross_product)
    return area

#判断是否属于水印部分
def is_watermar_area(polygon1, polygon2=[[137.0, 179.0], [458.0, 187.0], [457.0, 234.0], [136.0, 227.0]], gate=0.1) -> bool:
    #创建两个Polygon对象
    poly1 = Polygon(polygon1)
    poly2 = Polygon(polygon2)

    #计算两个多边形的交集
    intersection = poly1.intersection(poly2)

    #计算交集的面积
    intersection_area = intersection.area

    #计算两个多边形的并集面积
    union_area = poly1.area + poly2.area - intersection_area

    #计算交并比
    iou = intersection_area / union_area
    print("iou: ", iou)

    return iou > gate, iou
```




Dataset Cleaning



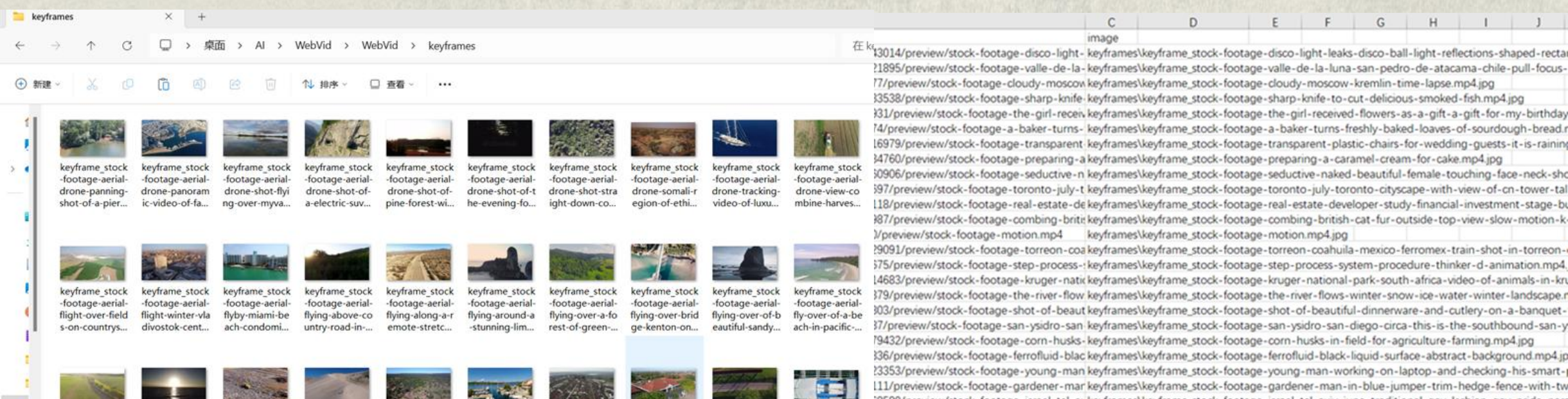
Dataset

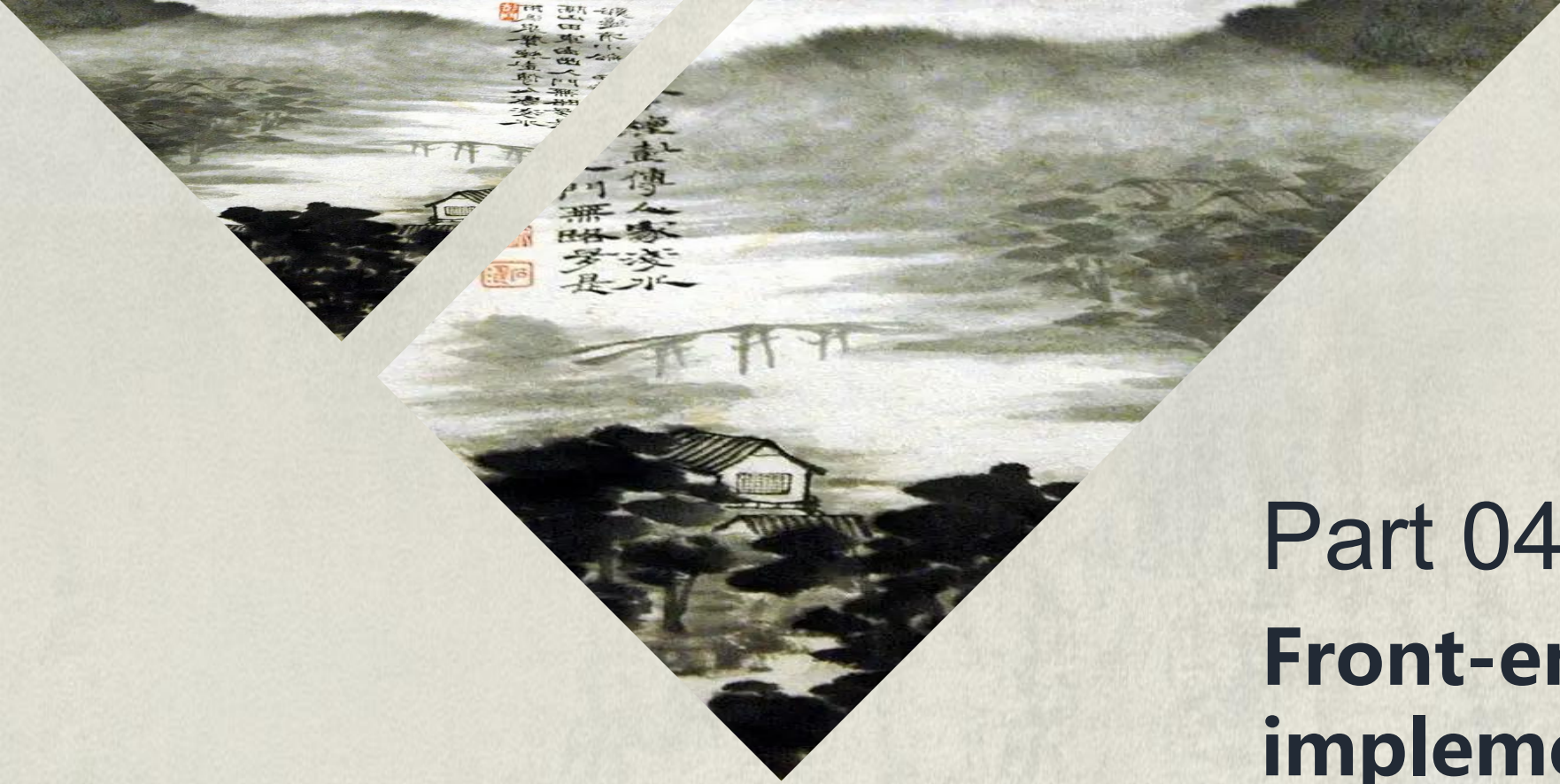
Firstly 5000 data from WebVid dataset were selected as video sets, which need to be cleaned before pairing with images.

The following two scoring criteria are selected in this dataset to clean the video set:

1. optic flow score: detect the optical flow, if the change between two frames is bigger, this optical flow score will also be higher for detecting the size of motion changes in the video;
2. aesthetic score: judge the aesthetic degree of the video itself. According to the actual situation and the paper reference to set the threshold, aesthetic score: 0.5 and optical flow score: 0.2

After cleaning, use Opencv to extract the first frame of the video from the video set to get its corresponding picture, and write it to a new csv file to form a video-picture pair dataset. Remove the data where the first frame is a solid color.





Part 04

Front-end and back-end implementations

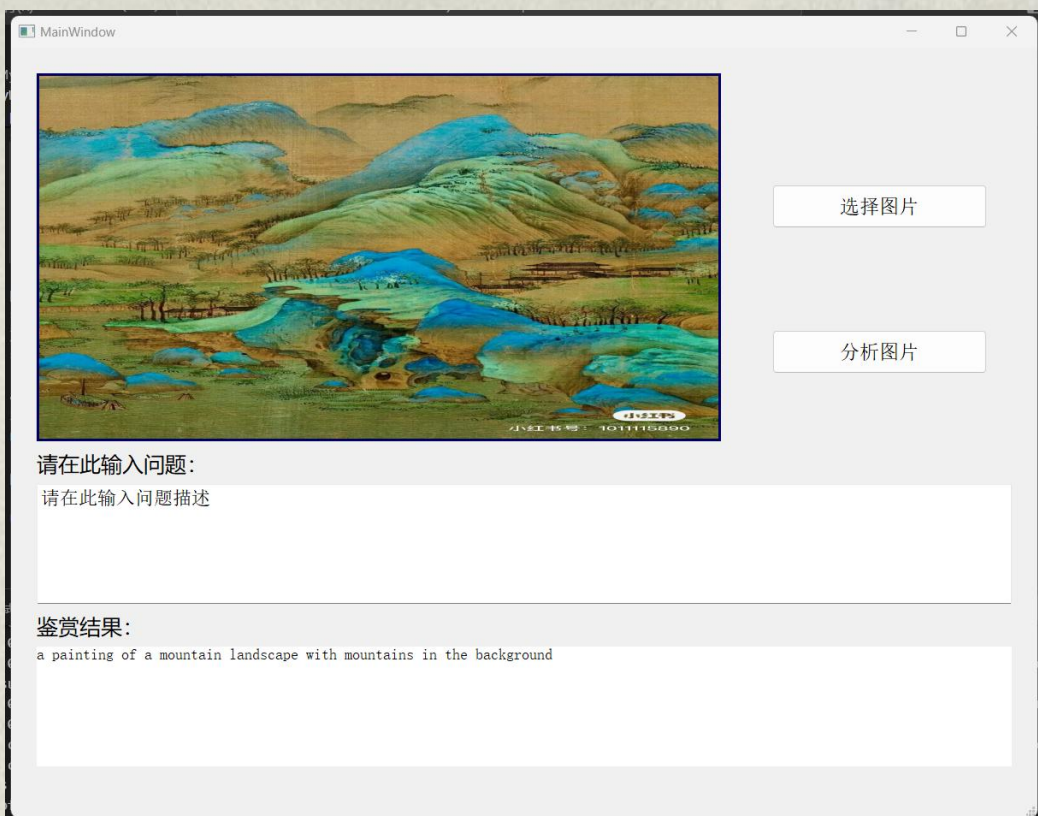


PYQT+SOCKET



Preliminary

- We use PYQT as the front-end interface, and also use Socket TCP to communicate with the server, send image files, and accept the returned results.
- However, due to unstable connection, cumbersome operation, interface modification is more difficult, so we chose a more lightweight, beautiful, more professional! Gradio



```
def deal_data(conn, addr):
    # 使用split()方法来分割文件路径
    file_name = filename.split("/")[-1]
    filePath = dirPath + file_name

    received_size = 0
    f = open(filePath, 'wb')
    while received_size < file_size:
        size = 0 # 准确接收数据大小, 解决粘包
        #print(received_size)
        if file_size - received_size > 1024: # 多次接收
            size = 1024
        else: # 最后一次接收完毕
            size = file_size - received_size

        filedata = conn.recv(size) # 多次接收内容, 接收大数据
        filedata_len = len(filedata)
        received_size += filedata_len
        #print('已接收: ', int(received_size / file_size * 100), "%")
        f.write(filedata)
    f.close()
    print('接收完毕')

# time.sleep(2)
```

问题 15 输出 调试控制台 终端 窗口

```
(base) PS E:\Projects\Python> & C:/Users/61771/AppData/Local/Programs/Python/Python311/python.exe e:/Projects/Python/CV/sockettest/fileClient.py
服务器已连接
put E:/Projects/Python/CV/sockettest/test.jpg 1488498
已接收指令及文件名
返回消息: 已接收
(base) PS E:\Projects\Python>
```

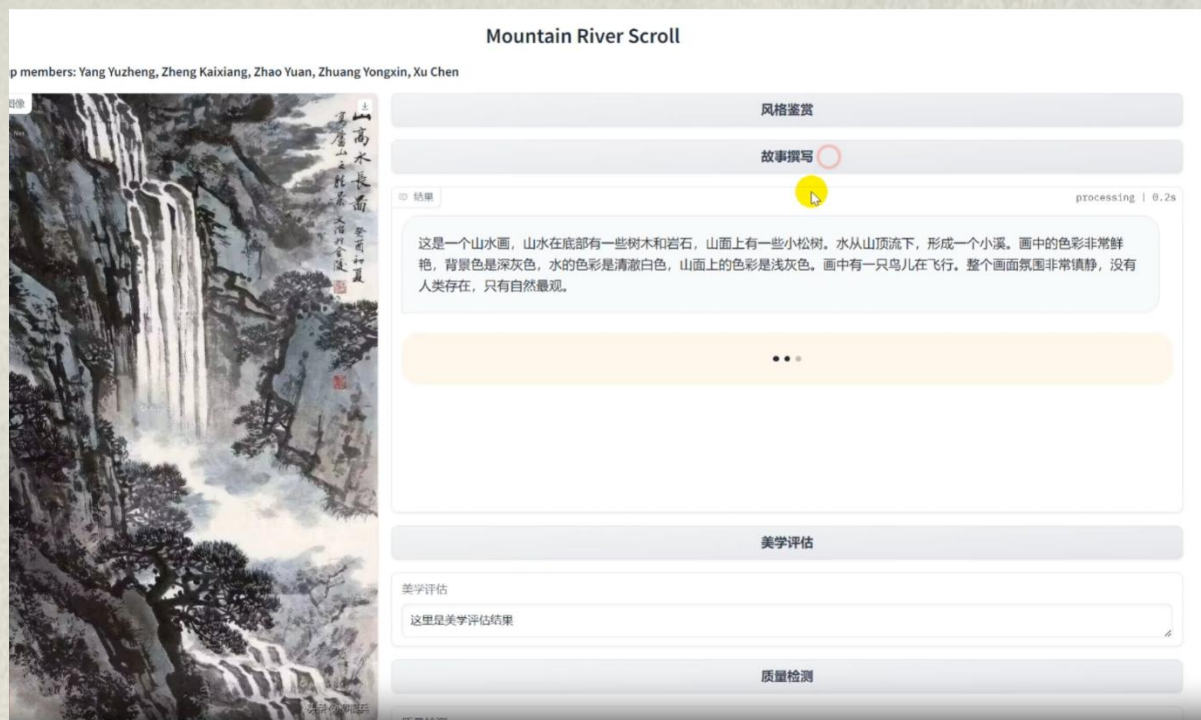



Gradio

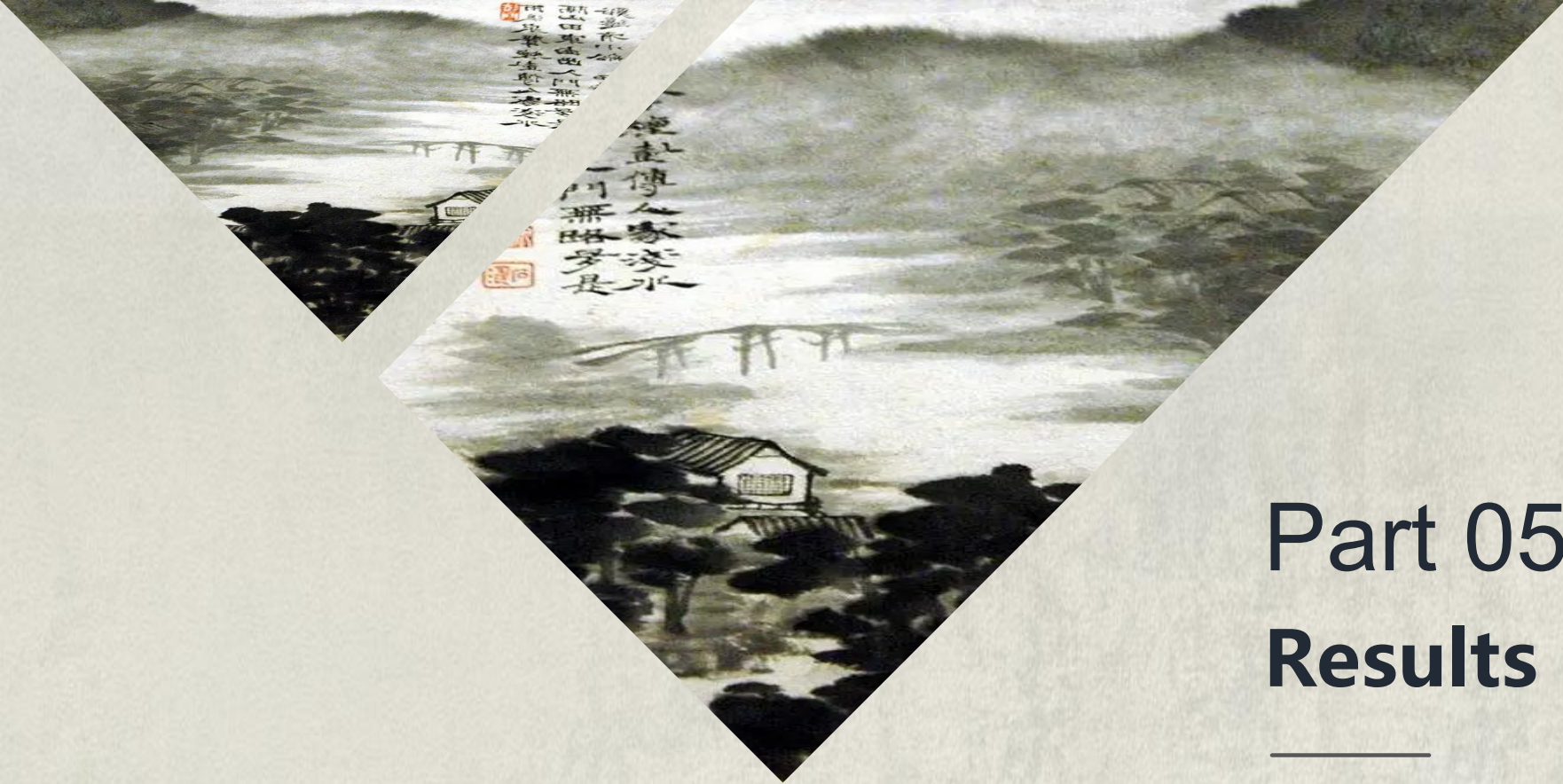


Improvements

- Use Gradio library to build the front-end interface, and bind links with the back-end model and so on, synchronize the initialization of the model and the interface at startup, you can directly open the display site locally.



```
def gradio_ask(prompt, chat_state, img_list, num_beams, temperature):  
    chatbot = gr.Chatbot(label='结果')  
    # 美学评估  
    bea_button = gr.Button(value="美学评估", interactive=False)  
    bea_text_out = gr.Textbox(label='美学评估', placeholder='这里是美学评估结果', interactive=False)  
    # 质量检测  
    quality_button = gr.Button(value="质量检测", interactive=False)  
    quality_text_out = gr.Textbox(label='质量检测', placeholder='这里是质量结果', interactive=False)  
  
    upload_button.click(upload_img, [image, chat_state], [image, appreciation_button, upload_button, bea_button, quality_button])  
    appreciation_button.click(gradio_ask, \n    [ chatbot, chat_state], [ chatbot, chat_state]).then(  
        gradio_answer, [chatbot, chat_state, img_list, num_beams, temperature], \n        [chatbot, chat_state, img_list, story_button]  
    )  
    story_button.click(storyWrite, [chatbot], [chatbot])  
  
clear.click(gradio_reset, [chat state, img list], [chatbot, image, appreciation button, upload but
```

Part 05

Results Showcase



Video Demon



Gradio

127.0.0.1:7860

知 首页 · 知乎 译 百度翻译 · 200种语... 哔哩哔哩 CSDN 展带教务平台 学生选课 GitHub Gitee.com 超级简历WonderC... 学习 展带 素材 游戏 视频/磁力 AI 梯子 招聘 竞赛 vr大空间

Mountain River Scroll

Group members: Yang Yuzheng, Zheng Kaixiang, Zhao Yuan, Zhuang Yongxin, Xu Chen

Image

↑

将图像拖放到此处

- 或 -

点击上传

↑

📷

📄

上传图片

重置

beam search numbers)

1

Temperature

1

风格鉴赏

故事撰写

结果

美学评估

质量检测

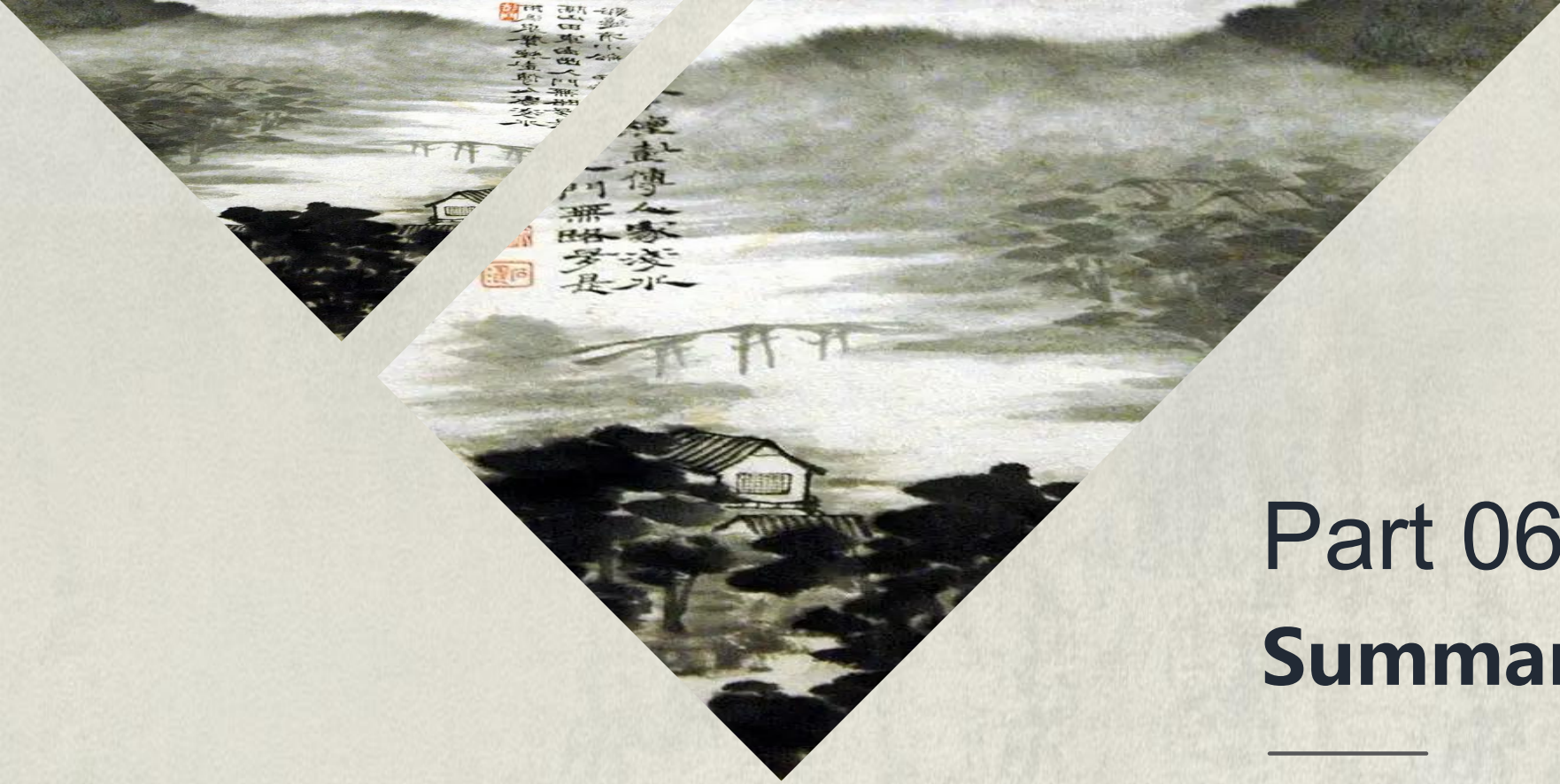
美学评估

这里是美学评估结果

质量检测

搜索

23:11 2024/6/19



Part 06

Summary and Outlook



Summary and Outlook



In the project, we used several large models in series and parallel to realize the description, interpretation and aesthetic assessment for the national style images. However, there are still some shortcomings.

- We have realized the integration of multi-model deployment, and built a more complete project of Chinese painting appreciation and animation generation.
- By combining Chinese landscape painting with AI technology, it injects innovative artistic expression into traditional Chinese painting, expands the boundaries of Chinese painting art, and promotes the dissemination and exchange of traditional culture.
 - In model invocation, non-locally deployed models, after multiple uses, fail to connect to the source URL due to network fluctuations, resulting in model invocation failure.
 - The algorithms used in data cleansing were not fully reproduced, only a portion of the algorithms were used to analyze the dataset with an accuracy that could be improved.
 - The effect of Image to Video Generation is relatively poor.



Division Of Labor

➤ Yang YuZheng

- Model exploration and deployment
- Algorithm improvement analysis
- Project integration

➤ Zheng Kaixiang

- Model exploration and deployment
- Thesis study
- Front and rear end construction

➤ Zhuang Yongxin

- Model exploration and deployment
- Thesis study

➤ Zhao Yuan

- Model exploration
- Dataset collection and processing
- Thesis study

➤ XV Chen

- Dataset collection and processing
- Thesis study





Thanks

